Worst case of each method: how much worse than the best method on the same case, % of the reference value.

General weather gaps, random and event placements, each event week counted once. Darker = larger (log scale).

Below the line: the nearby station alone, scored only on gaps where it passed its checks, so not comparable.

a Annual heating: under 7% for all methods

	3 hours	12 hours	48 hours	1 week
Stepwise	**0.02**	**0.26**	**1.2**	**5.2**
Stepwise, no switch	0.02	0.26	1.2	5.3
Stepwise, patterns first	0.02	0.58	1.4	5.3
Interpolation then patterns	0.02	0.58	1.4	–
ERA5 only	0.10	0.47	0.98	5.2
Four-week average	0.02	1.2	3.1	5.2
Linear interpolation only	0.02	0.21	1.6	6.5
Nearby station (own subset)	0.03	0.17	0.55	0.54

b Annual cooling: linear interpolation worst

	3 hours	12 hours	48 hours	1 week
Stepwise	**1.2**	**2.3**	**6.4**	**11**
Stepwise, no switch	1.2	2.3	6.4	14
Stepwise, patterns first	1.2	8.2	8.9	14
Interpolation then patterns	1.2	8.2	8.9	–
ERA5 only	0.88	2.6	4.9	11
Four-week average	1.2	20	27	11
Linear interpolation only	1.2	12	21	54
Nearby station (own subset)	0.51	1.5	2.6	14

c Heating peak: no switch doubles the worst case

	3 hours	12 hours	48 hours	1 week
Stepwise	**3.0**	**7.5**	**14**	**14**
Stepwise, no switch	3.0	7.5	14	31
Stepwise, patterns first	3.0	10	14	31
Interpolation then patterns	3.0	10	14	–
ERA5 only	10	11	17	14
Four-week average	3.0	11	15	14
Linear interpolation only	3.0	6.9	11	15
Nearby station (own subset)	4.4	2.7	2.8	7.0

d Cooling peak: no switch reaches 85%

	3 hours	12 hours	48 hours	1 week
Stepwise	**8.6**	**12**	**20**	**16**
Stepwise, no switch	8.6	12	20	85
Stepwise, patterns first	8.6	12	23	85
Interpolation then patterns	8.6	12	23	–
ERA5 only	23	16	16	16
Four-week average	8.6	30	38	16
Linear interpolation only	8.6	25	14	38
Nearby station (own subset)	2.2	5.9	11	12